Peng Liu, Ph.D.

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Boston, MA, USA

Research Interests

Automated proofreading and error correction in connectomics (ML approaches for error detection and human-in-the-loop proofreading)

Label-efficient biomedical image segmentation (semi-supervised and weakly-supervised learning)

Foundation models and benchmarks for scalable multi-organelle EM segmentation

Academic Appointments and Professional Experience

Academic Appointments

Aug 2024 – Present **Boston College, Department of Computer Science** **Boston, MA, USA**
Postdoctoral Researcher (Advisor: Prof. Donglai Wei)

- Research focus: 3D volume EM segmentation for mitochondria, nuclei, and pan-organelle analysis.
- Led benchmark-centered research including MitoEM 2.0 (*Scientific Data*, under review).
- **1st Place**, CellMap Segmentation Challenge (2026; eFIB-SEM, 40+ classes).
- Ongoing work on skeleton-guided learning and multi-resolution mixture-of-experts models for instance segmentation.

Industry Experience

Jul 2017 – Jul 2019 **Meituan** **Beijing, China**
Search and Recommendation Algorithm Engineer

- Built ranking and recommendation models for travel products to improve CTR and CVR.
- Core methods: Graph Embedding, DNN, and Wide&Deep.

Education

2019 – 2024 **Shanghai Jiao Tong University** **Shanghai, China**
Ph.D. in Biomedical Engineering

- **Advisor**: Guoyan Zheng
- **Dissertation**: Semi-Supervised Medical Image Analysis Based on Consistency Regularization

2015 – 2018 **University of Chinese Academy of Sciences (Aerospace Information Research Institute)** **Beijing, China**
M.Sc. in Signal and Information Processing

- **Thesis**: Object Detection in High-Resolution Remote Sensing Images Based on Deep Learning

2011 – 2015 **China University of Geosciences** **Beijing, China**
B.Sc. in Geographic Information System

Publications (∗ indicates equal contribution)

Journal Articles

1. **P. Liu** and G. Zheng, "Handling imbalanced data: Uncertainty-guided virtual adversarial training with batch nuclear-norm optimization for semi-supervised medical image classification," *IEEE Journal of Biomedical and Health Informatics*, vol. 26, no. 7, pp. 2983–2994, 2022, **(JCR Q1, CAS Q1, IF=7.7)**.
2. **P. Liu** and G. Zheng, "Cvcl: Context-aware voxel-wise contrastive learning for label-efficient multi-organ segmentation," *Computers in Biology and Medicine*, vol. 160, p. 106 995, 2023, **(JCR Q1, IF=7.7)**.
3. C. Chen*, **P. Liu***, and Y. Song, "Diagnostic performance for severity grading of hip osteoarthritis and osteonecrosis of femoral head on radiographs: Deep learning model vs. board-certified orthopaedic surgeons," *Osteoarthritis Imaging*, vol. 3, no. 2, p. 100 092, 2023.
4. **P. Liu**, M. Zhang, X. Gao, B. Li, and G. Zheng, "Joint lymphoma lesion segmentation and prognosis prediction from baseline fdg-pet images via multitask convolutional neural networks," *IEEE Access*, vol. 10, pp. 81 612–81 623, 2022, **(JCR Q2, IF=4.8)**.
5. Z. Li, K. Chen, **P. Liu**, X. Chen, and G. Zheng, "Entropy and distance maps-guided segmentation of articular cartilage: Data from the osteoarthritis initiative," *International Journal of Computer Assisted Radiology and Surgery*, pp. 1–8, 2022, **(JCR Q2, IF=2.7)**.
6. **P. Liu** and G. Zheng, "Cr-gloco: Cross-resolution learning via global-local context consistency for semi-supervised 3d medical segmentation," *Computerized Medical Imaging and Graphics*, 2026, Accepted and available online (in press), **(JCR Q1, IF=4.9)**.

Preprints

1. A. Bhardwaj, C. W. Dell, M. R. Mikolaj, H. Spiers, A. Harned, B. Kuppusamy, **P. Liu**, D. Wei, E. Sterneck, and K. Narayan, "Nucleonet and dropnet: Generalist deep learning models for instance segmentation of nuclei and lipid droplets from electron microscopy images," *bioRxiv*, p. 2026.04.02.713930, 2026. DOI: `10.64898/2026.04.02.713930`.
2. M. Wang, **P. Liu**, Y. Zhao, B. Wang, J. Wan, L. Nie, and D. Wei, "Nugraph: Graph-based reasoning over 3d primitives for nucleus segmentation correction," *bioRxiv*, p. 2026.05.16.725603, 2026, Under review at IEEE Transactions on Medical Imaging. DOI: `10.64898/2026.05.16.725603`.

Conference Proceedings

1. **P. Liu** and G. Zheng, "Semi-supervised learning regularized by adversarial perturbation and diversity maximization," in *Machine Learning in Medical Imaging: 12th International Workshop, MLMI 2021, Held in Conjunction with MICCAI 2021*, Springer, 2021, pp. 199–208, **(EI)**.
2. **P. Liu** and G. Zheng, "Context-aware voxel-wise contrastive learning for label efficient multi-organ segmentation," in *International Conference on Medical Image Computing and Computer-Assisted Intervention*, Springer, 2022, pp. 653–662, **(EI, CCF B)**.
3. Q. Zhou*, **P. Liu***, and G. Zheng, "Context-aware cutmix is all you need for universal organ and cancer segmentation," in *Fast, Low-Resource, and Accurate Organ and Pan-Cancer Segmentation in Abdomen CT*, Springer, 2023, **(EI)**.
4. Q. Zhou, **P. Liu**, and G. Zheng, "Partially supervised multi-organ segmentation via affinity-aware consistency learning and cross site feature alignment," in *International Conference on Medical Image Computing and Computer-Assisted Intervention*, Springer, 2023, pp. 671–680.

Under Review

1. **P. Liu**, B. Shen, L. Liu, Q. Wang, S. Zhang, A. Bhardwaj, I. Arganda-Carreras, K. Narayan, and D. Wei, "Mitoem 2.0: A benchmark for challenging 3d mitochondria instance segmentation from em images," Minor Revision at Scientific Data, 2025, **(JCR Q1, IF=6.9)**.
2. D. Shi, Y. Hou, Y. Yan, T.-h. Zhang, W. C. Joesten, **P. Liu**, Y. Wang, M. Gautam, J. Lim, L. Zheng, J. Gould, B. Ko, X. Niu, M.-C. Cheng, J.-C. Hsieh, F. Levet, D. Cai, A. Draelos, D. J. Cai, D. Wei, and C. Linghu, "Simultaneous brain-wide single-cell recording resolves spatiotemporal memory architecture," **Under Review at Nature**, 2026. url: `https://doi.org/10.1101/2026.05.21.726120`.

Manuscripts in Preparation

1. **P. Liu** and D. Wei, *Anchorslot: Hybrid target coding for volumetric em organelle segmentation*, To be submitted to IEEE Transactions on Medical Imaging, 2026.
2. **P. Liu**, D. Roy, and D. Wei, *Affinity-guided superpoint learning for false-merge correction in 3d em mitochondria segmentation*, To be submitted to IEEE Transactions on Medical Imaging, 2026.

Open-Source Software

CellAble	github.com/pliu-ai/cellable Interactive 2D/3D annotation and human-in-the-loop proofreading tool for automated cell reconstructions in electron microscopy. Supports TIFF stack viewing, AI-assisted segmentation, manual mask editing, connected-component analysis, and 3D VTK rendering. Built with Python, PyQt5, and VTK.

Honors and Awards

Research Competitions

2026	1^{st} **Place**, CellMap Segmentation Challenge (eFIB-SEM multi-organelle segmentation, 40+ classes).
2023	2^{nd} **Place**, MICCAI FLARE 2023 Challenge (abdominal organ and pan-tumor segmentation; leveraged label-efficient and semi-supervised learning for automated annotation at scale across 4000+ CT scans).
2018	**Ranking: 20/120**, MDD Cup 2018 of Meituan Group. In Kaggle MDD Cup 2018.
2017	**Ranking: 7/148**, MDD Cup 2017 of Meituan Group. In Kaggle MDD Cup 2017.

Scholarships

2012	**National Encouragement Scholarship**. Awarded by Ministry of Education of China.
2012, 2013, 2014	**University Scholarships**. Awarded by China University of Geosciences.

Teaching Experience

Teaching Assistant

Shanghai Jiao Tong University, CS 2901: Programming Ideas and Methods (C), Fall 2021 and Fall 2022.

Shanghai Jiao Tong University, CS 170: Programming Ideas and Methods (C), Fall 2019 and Fall 2020.

Courses Prepared to Teach

Programming Fundamentals (C/C++/Python)

Data Structures and Algorithms

Machine Learning and Deep Learning

Medical Image Processing and Analysis

Professional Service

Journals: *IEEE Transactions on Medical Imaging*, *Medical Image Analysis*, *IEEE Journal of Biomedical and Health Informatics*, *Neural Networks*, *Computerized Medical Imaging and Graphics*

Conferences: MICCAI 2021, 2022

Patents and Software

Patent

Peng Liu; Guoyan Zheng. Multi-organ segmentation method, training method, medium, and electronic device. Chinese invention patent application No. 202211185528.1, filed on September 27, 2022.

Software Copyright

Jiale Wang; Peng Liu; Guoyan Zheng. Deep-learning-based Spine Localization and Segmentation System (Spine_loc_seg) V1.0. Registration No. 2023SR0776267, registered on April 28, 2023.

Technical Skills

Languages	English (professional proficiency; TOEFL, CET-6), Mandarin Chinese (native).
Coding	Linux (7 years), C++ (10 years), Python (7 years), PyTorch (6 years), Git.